

Analysis of H-Plane Waveguide Components with Dielectric Obstacles by the BI-RME Method

Paolo Arcioni¹, Vicente Boria², Maurizio Bozzi¹, Giuseppe Conciauro¹, and Benito Gimeno³

¹ University of Pavia, Department of Electronics, Via Ferrata 1, I-27100 Pavia, Italy
Phone: +39 0382505200, fax: +39 0382422583
e-mail: arcioni@ele.unipv.it, bozzi@ele.unipv.it, conciauro@ele.unipv.it

² Univ. Polit cnica de Valencia, Dpto. Comunicaciones, Camino de Vera s/n, E-46022, Valencia, Spain
e-mail: vboria@dcom.upv.es

³ Univ. de Valencia, Dpto. F sica Aplicada, Dr. Moliner 50, E-46100 Burjassot, Valencia, Spain
e-mail: benito.gimeno@uv.es

Abstract—This paper presents a novel method for the wideband modeling of H-plane waveguide components, which include dielectric obstacles. Both the planar component and the dielectric obstacles have an arbitrary shape. This method is based on the combination of the segmentation technique and the Boundary Integral-Resonant Mode Expansion (BI-RME) method. Two examples show the effectiveness of the method.

I. INTRODUCTION

Dielectric resonators made of high permittivity material have found practical applications in microwave and millimeter-wave components due mainly to their small size, low cost, good temperature stability, and high-Q values. In the last decades, many researchers have made a very important effort in the characterization of dielectric resonators [1]–[6] and dielectric obstacles inside rectangular waveguides [7]–[12]. In particular, the design of band-pass or band-stop filters were considered in [7] and a tunable band-stop filter was presented in [9]. Few of the available results found in the technical literature, however, can be used for a full-wave filter design, since they are essentially concerned with the determination of the scattering properties of isolated obstacles inside waveguides [9], [12], or the characterization of full-height circular dielectric posts by lumped networks [8], [10], [11]. A full-wave characterization of these structures is therefore crucial to develop efficient CAD tools for the analysis and design of filters based on dielectric resonators.

In this paper we present a novel accurate and computationally efficient technique for the full-wave analysis of H-plane waveguide structures including dielectric obstacles with arbitrary profile. The method is based on a segmentation technique, which leads to consider a number of homogeneously filled subcircuits, each one characterized by its General-

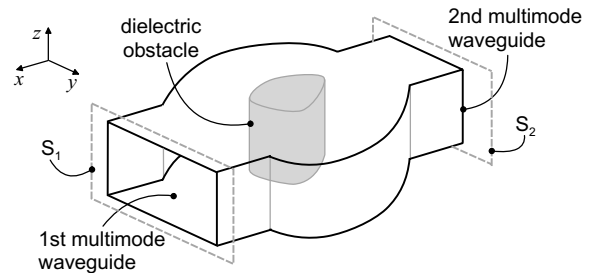


Fig. 1. H-plane waveguide component including a dielectric obstacle: both the circuit and the obstacle may have an arbitrary cross-section.

ized Admittance Matrix (GAM). These matrices are obtained in the form of a pole expansion in the frequency domain by using the technique described in [13], Chapters 2 and 6. The frequency response of the whole structure is obtained by combining the GAMs of the subcircuits according to the algorithm described in [14], which yields the resulting GAM still in the same form as the GAMs of the subcircuits. The algorithm can handle waveguide structures with an arbitrary shape, thus permitting a complete flexibility in the design. Its numerical efficiency allows to perform the wide-band analysis of the structure in very short computing times, that are compatible with optimization procedures. Comparisons with results from the technical literature fully validate the proposed technique.

II. OUTLINE OF THE THEORY

Let us consider an H-plane waveguide component with an arbitrary shape, including, for the sake of simplicity, a single dielectric obstacle with arbitrary cross-section (Fig. 1). The extension to the case of multiple obstacles is straightforward. We assume that the tangential field at the reference plane S_w is due to the combination of a number of TE_{n0} modes ($n = 1, \dots, N_w$): therefore, the field inside the structure is z -independent, and the problem is essentially two-dimensional. The component is characterized through its GAM, which relates modal currents and voltages at the reference planes $S_1, S_2, \dots, S_W$. By using the segmentation technique, the inhomogeneous H-plane component (Fig. 2a) is divided into two homogeneous subcircuits, one filled with air (Fig. 2b), the other filled with dielectric (Fig. 2c). Each subcircuit is analyzed separately, and characterized by its GAM. In particular, when calculating the GAM of the air-filled subcircuit, we must consider, besides the reference planes $S_1, S_2, \dots, S_W$, an additional “reference surface” S_I , corresponding to the interface between the two subcircuits (see Fig. 2b). On S_w the transverse electric and magnetic fields are expressed in the form

$$\vec{E}_w = \sum_{n=1}^{N_w} V_w^{(n)} \vec{e}_w^{(n)} \quad (1)$$

$$\vec{H}_w = \sum_{n=1}^{N_w} I_w^{(n)} \vec{h}_w^{(n)} \quad (2)$$

where $\vec{e}_w^{(n)}$ and $\vec{h}_w^{(n)}$ are the electric and magnetic modal vectors, respectively, of the TE_{n0} modes of the w -th waveguide, and $V_w^{(n)}$ and $I_w^{(n)}$ are the corresponding modal voltages and currents. On S_I , the transverse electric and magnetic fields are expressed in the form

$$\vec{E}_I = \sum_{m=1}^M V_m f_m \vec{u}_z \quad (3)$$

$$\vec{H}_I = \sum_{m=1}^M I_m f_m \vec{n} \times \vec{u}_z \quad (4)$$

where f_m are z -independent scalar basis functions defined on S_I , V_m and I_m are the corresponding “voltages” and “currents”, $\vec{n}$ is the inward normal on the inner boundary, and $\vec{u}_z$ is the unit vector in the z direction (see Fig. 1). As a consequence, the GAM of the air-filled subcircuit relates the currents $I_w^{(n)}$, I_m to the voltages $V_w^{(n)}$, V_m , corresponding

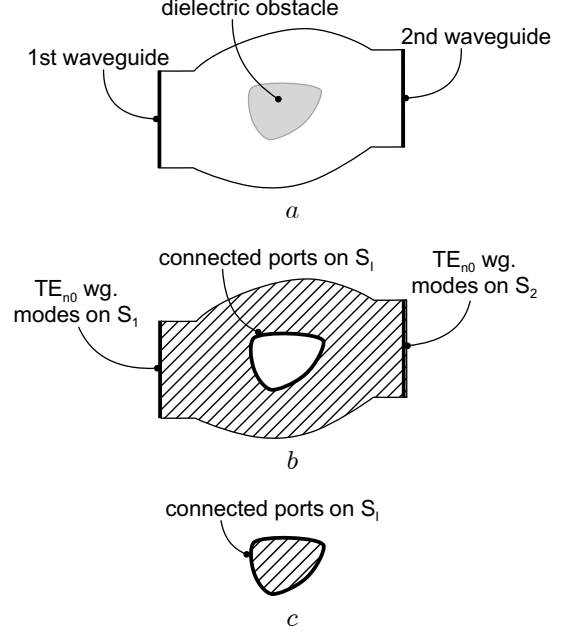


Fig. 2. Segmentation of the waveguide components: (a) cross-section of the H-plane component; (b) cross-section of the air-filled subcircuit; (c) cross-section of the dielectric-filled subcircuit.

to the “external ports” (defined on $S_1, S_2, \dots, S_W$) and to the “connected ports” (defined on S_I).

Since the air-filled subcircuit is homogeneous, the Boundary Integral-Resonant Mode Expansion (BI-RME) method [13] can be used for the calculation of the GAM in the form of a pole-expansion in the frequency domain [15], [16]

$$\mathbf{Y}(\omega) = \frac{\mathbf{A}}{j\omega} + j\omega\mathbf{B} + j\omega^3 \sum_{i=1}^M \frac{\mathbf{c}^{(i)} \mathbf{c}_T^{(i)}}{\omega_i^2 - \omega^2} \quad (5)$$

In (5) ω_i are the resonant frequencies of the first resonating modes of the cavity obtained by short-circuiting all the ports (poles of the GAM). As discussed in [13], [15], to obtain a good accuracy in a frequency band from 0 to ω_{max} , it is sufficient to include in the pole expansion only the poles up to a frequency two or three times ω_{max} . Moreover, matrices $\mathbf{A}$, $\mathbf{B}$ and vectors $\mathbf{c}^{(i)}$ are real and independent of the frequency. All these quantities can be efficiently calculated by using the algorithm described in [15], [16], which can be extended to take into account the presence of the ports defined on S_I . This extension requires only minor modifications of the algorithm, and is not reported here for brevity.

A similar expression is obtained for the GAM of the dielectric-filled subcircuit. The electric and magnetic fields on S_I —that, in this case, represents the lateral wall of the dielectric obstacle (Fig. 2c)—are expressed in the same form as in (3) and (4),

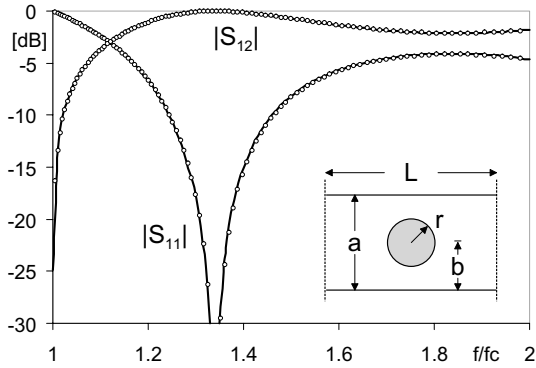


Fig. 3. Frequency response of a centered dielectric post ($r/a = 0.25$, $b/a = 0.5$, $\epsilon_r = 4$) inside a rectangular waveguide ($L/a = 1.5$). Scattering parameters refer to the TE_{10} mode; frequency is normalized to the cutoff frequency of that mode. Solid lines: this method; dots: results by the method [17].

and the GAM of this subcircuit relates the currents I_m to the voltages V_m . Also in this case, the circuit is homogeneous, and its GAM can be deduced in the form (5) by using the above mentioned algorithm.

Once the GAMs of the two subcircuits have been determined, the GAM of the whole component is obtained by enforcing the continuity of the tangential electric and magnetic fields on S_I , *i.e.*, the continuity of the voltages and the currents on the connected ports. A very efficient method to determine the resulting GAM has been introduced in [14]: using this algorithm the GAM of the whole component is obtained still in the same form of a pole expansion as the GAM of the subcircuits.

As evidenced in the following examples, the main advantage of the proposed method is its numerical efficiency: very short computing times are needed to compute the resonant frequencies, the matrices $\mathbf{A}$, $\mathbf{B}$, and the vectors $\mathbf{c}^{(i)}$ of the two subcircuits, and to deduce the resulting GAM in the form (5). Once the pole expansion of the component is known, its frequency response (in terms of admittance or scattering parameters) can be deduced at any frequency up to ω_{max} , without any substantial computational effort.

III. EXAMPLES

In the implementation of the method we approximate the boundary of the dielectric obstacle by a polygonal, and we use parabolic splines as basis functions f_m (see [13] for more detail).

The first validation example refers to a centered

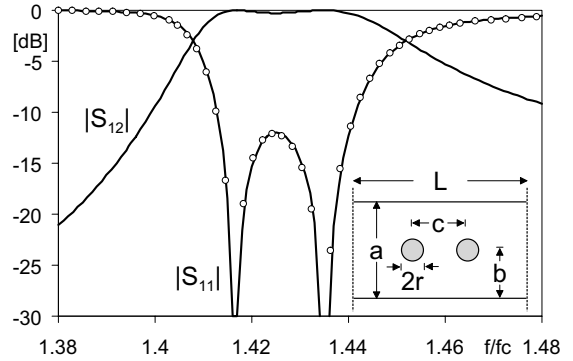


Fig. 4. Frequency response of the filtering structure considered in [12]: $r/a = 0.05$, $b/a = 0.5$, $c/a = 0.4$, $\epsilon_r = 112.5$, $L/a = 1.4$. Scattering parameters refer to the TE_{10} mode; frequency is normalized to the cutoff frequency of that mode. Solid lines: this method; dots: results by [11].

cylindrical post with rather low dielectric constant ($\epsilon_r = 4$), placed inside a rectangular waveguide. Fig. 3 shows the excellent agreement between the frequency response of the structure calculated by the present method and the one obtained by a different approach [17]. We used 16 basis functions to represent the fields on the boundary of the dielectric post, and we included in the pole expansion of the GAMs 21 and 12 poles in the case of the air- and dielectric-filled subcircuits, respectively. Three modes were considered on each of the waveguide reference sections. The computation time for deriving the GAMs of the subcircuit and for calculating the resulting GAM was only 2 sec on a PC Pentium III @ 800 MHz.

We considered also the filtering structure shown in the inset of Fig. 4, consisting of two centered dielectric posts of very high dielectric constant ($\epsilon_r = 112.5$) inside a rectangular waveguide. This configuration has been studied in [11] by a combination of the finite and the boundary element method (CFBEM). The response of the structure calculated by the present method and by the CFBEM is reported in Fig. 4, which shows a perfect agreement between the two methods. In our analysis we used 12 basis functions, and included 23 and 20 poles in the pole expansions of the GAMs of the air- and dielectric-filled subcircuits, respectively. Three modes were considered on each of the waveguide reference sections. Also in this case the method proved its numerical efficiency, since the computing time was only 3 sec for the complete wideband analysis of the structure.

IV. CONCLUSIONS

This paper presents a novel approach for the analysis of planar waveguide components including dielectric obstacles. The approach is based on the segmentation technique and the BI-RME method, and provides a wideband representation of the admittance matrix of the component. The reported examples demonstrate the accuracy and numerical efficiency of the method, showing that the typical computing times are well-suited to an optimization process. Therefore, this method can be used as a full-wave solver for the CAD of this class of components.

V. ACKNOWLEDGEMENT

This work was supported by the European Commission under the Research and Training Networks Programme, Contract No. HPRN-CT-2000-00043.

REFERENCES

- [1] J. Krupka, "Optimization of an Electrodynamical Basis for Determination of the Resonant Frequencies of Microwave Cavities Partially Filled with a Dielectric," *IEEE Trans. Microwave Theory Tech.*, Vol. MTT-31, No. 3, pp. 302-305, March 1983.
- [2] D. Kajfez and P. Guillon, *Dielectric Resonators*, Artech House, Inc., 1986.
- [3] P. H. Harms, Jin-Fa Lee, and R. Mittra, "A Study of the Nonorthogonal FDTD Method Versus the Conventional FDTD Technique for Computing Resonant Frequencies of Cylindrical Cavities," *IEEE Trans. Microwave Theory Tech.*, Vol. MTT-40, No. 4, pp. 741-746, April 1992.
- [4] J.A. Pereda, L.A. Vielva, A. Vegas, and A. Prieto, "Computation of Resonant Frequencies and Quality Factors of Open Dielectric Resonators by a Combination of the Finite-Difference Time Domain (FDTD) and Prony's Methods," *IEEE Microwave and Guided Wave Letters*, Vol. 2, No. 11, pp. 431-433, Nov. 1992.
- [5] N. Kaneda, B. Houshmand, and T. Itoh, "FDTD Analysis of Dielectric Resonators with Curved Surfaces," *IEEE Trans. Microwave Theory Tech.*, Vol. MTT-45, No. 9, pp. 1645-1649, Sept. 1997.
- [6] A. R. Weley and A. S. Mohan, "Microwave Filters with Improved Spurious Performance Based on Sandwiched Conductor Dielectric Resonators," *IEEE Trans. Microwave Theory Tech.*, Vol. MTT-49, No. 8, pp. 1501-1507, Aug. 2001.
- [7] J. N. Sahalos and E. Vafiadis, "On the Narrow-Band Microwave Filter Design Using Dielectric Rod," *IEEE Trans. Microwave Theory Tech.*, Vol. MTT-33, No. 11, pp. 1165-1171, Nov. 1985.
- [8] Chung-I G. Hsu and H. A. Auda, "Multiple Dielectric Posts in a Rectangular Waveguide," *IEEE Trans. Microwave Theory Tech.*, Vol. MTT-34, No. 8, pp. 883-891, August 1986.
- [9] R. Gesche and N. Loechel, "Scattering by a Lossy Dielectric Cylinder in a rectangular Waveguide," *IEEE Trans. Microwave Theory Tech.*, Vol. MTT-36, No. 1, pp. 137-144, Jan. 1988.
- [10] Chung-I G. Hsu and Hesaham A. Auda, "On the Realizability of the Impedance Matrix for Lossy Dielectric Posts in a rectangular Waveguide," *IEEE Trans. Microwave Theory Tech.*, Vol. MTT-36, No. 4, pp. 763-765, April 1988.
- [11] K. Ise and M. Koshiba, "Equivalent Circuits for Dielectric Posts in a Rectangular Waveguide," *IEEE Trans. Microwave Theory Tech.*, Vol. MTT-37, No. 11, pp. 1823-1825, Nov. 1989.
- [12] K. Siakavara and J. N. Sahalos, "The Discontinuity Problem of a Rectangular Dielectric Post in a Rectangular Waveguide," *IEEE Trans. Microwave Theory Tech.*, Vol. MTT-39, No. 9, pp. 1617-1622, Sept. 1991.
- [13] G. Conciauro, M. Guglielmi, and R. Sorrentino, *Advanced Modal Analysis*, John Wiley and Sons, 2000.
- [14] P. Arcioni and G. Conciauro, "Combination of Generalized Admittance Matrices in the Form of Pole Expansions," *IEEE Trans. Microwave Theory Tech.*, Vol. MTT-47, No. 10, pp. 1990-1996, Oct. 1999.
- [15] G. Conciauro, P. Arcioni, M. Bressan, and L. Perregrini, "Wideband Modeling of Arbitrarily Shaped H-Plane Waveguide Components by the 'Boundary Integral-Resonant Mode Expansion Method'," *IEEE Trans. Microwave Theory Tech.*, Vol. MTT-44, No. 7, pp. 1057-1066, July 1996.
- [16] P. Arcioni, M. Bressan, G. Conciauro, and L. Perregrini, "Generalized Y-Matrix of Arbitrary H-Plane Waveguide Junctions by the BI-RME Method," *International Microwave Symposium Digest (IMS1997)*, pp. 211-214, 1997.
- [17] H. Esteban, S. Cogollos, V.E. Boria, A. San Blas, and M. Ferrando, "A New Hybrid Mode Matching/Numerical Method for the Analysis of Arbitrarily Shaped Inductive Obstacles and Discontinuities in Rectangular Waveguides," *IEEE Trans. Microwave Theory Tech.*, Vol. MTT-50, No. 4, pp. 1219-1224, Apr. 2002.